

Supporting Information for:

Future inorganic nutrient and plankton dynamics in a temperate lake

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11 **Supporting Information S1 – Model configuration and equations**

12 *Physical configuration*

13 The lake ecosystem model operates in a zero-dimensional physical configuration that consists of two
14 layers stacked on top of each other, representing the upper epilimnion and the bottom hypolimnion
15 (Fig. 1) and reflecting the so-called slab physics approximation (e.g. Fasham et al. 1990; To et al.
16 2024),. The depth of the epilimnion is known as the mixed layer depth (MLD) and varies seasonally.
17 The epilimnion is biologically active and is assumed to be well-mixed so that all model properties,
18 physical and biological, are considered homogeneously distributed within its volume. The
19 hypolimnion is biologically inactive. In the hypolimnion, a fixed amount of nutrient, N_0 , replenishes
20 through physical mixing the inorganic nutrient pool of the epilimnion, N . Changes in motile
21 (zooplankton) and non-motile (nutrient, detritus, and phytoplankton) model variables, induced by
22 physical processes, can occur due to (1) exchange of fluxes driven by diffusive mixing between the
23 epilimnion and hypolimnion and (2) variations in the MLD. The variation in zooplankton biomass
24 due to changes in the MLD is described by the following term (Evans and Parslow 1985):

$$26 \quad \lambda_z = \frac{h}{MLD(t)}, \quad (S1)$$

27
28 with $h = \frac{dMLD(t)}{dt}$. Here we assume that zooplankton can maintain their position within the upper
29 epilimnion. The shallowing or deepening of the MLD will, respectively, increase or decrease
30 zooplankton concentrations. A shallowing MLD leaves the concentrations of non-motile variables
31 (i.e. nutrient, phytoplankton, and detritus) unchanged because their reduction in number due to
32 detrainment is compensated by the reduction in volume of the epilimnion. A deepening MLD reduces
33 the concentrations of non-motile variables, unless N_0 is larger than N , that is, when the nutrient input
34 from the hypolimnion is greater than the nutrient concentration in the epilimnion. The non-motile
35 variables are also subject to diffusive mixing. The variations in biomasses of non-motile variables
36 due to physical processes can be written as follows (Evans and Parslow 1985):

$$37 \quad \lambda = \frac{\omega + h^+(t)}{MLD(t)}, \quad (S2)$$

38
39 with ω being the diffusive mixing constant and

$$40 \quad h^+(t) = \max(0, h). \quad (S3)$$

41

42 *Biological components*

43 The model consists of a set of 14 ordinary differential equations:

$$\frac{dN}{dt} = - \sum_i \mu_i A_i + (1-\gamma)\varepsilon \sum_i \sum_j G_{ij} Z_j + \varphi D + \lambda N_0 - \lambda N \quad (\text{S4})$$

$$\frac{dA_i}{dt} = (\mu_i - \phi_A - \lambda) A_i - \sum_i \sum_j G_{ij} Z_j \quad (\text{S5})$$

$$\frac{dZ_j}{dt} = \left(\varepsilon \gamma \sum_i \sum_j G_{ij} Z_j - \phi_Z - \lambda_Z - \eta_Z Z_j \right) Z_j \quad (\text{S6})$$

$$\frac{dD}{dt} = \sum_i \phi_P A_i + \sum_j \phi_Z Z_j + (1-\varepsilon) \sum_i \sum_j G_{ij} Z_j - (\varphi + \lambda) D \quad (\text{S7})$$

44

45 with $i = 1, \dots, 10$ and $j = 1, 2$. All the terms in the equations are described in the following.

46

47 The dynamics of nutrient (Eq. 4) is driven by nutrient lost to phytoplankton growth, $\sum_i \mu_i A_i$,
 48 and nutrient replenishment from (1) the portion of grazed materials that is not assimilated by
 49 zooplankton, $(1-\gamma)\varepsilon \sum_i \sum_j G_{ij} Z_j$, (2) the remineralization of detritus, φD , and (3) the exchange
 50 between the epilimnion and hypolimnion via vertical mixing, $\lambda(N_0 - N)$, where N_0 is the nutrient
 51 concentration in the hypolimnion. We determined the value N_0 based on average winter observations,
 52 when the entire water column is well mixed and nutrient concentrations are homogeneously
 53 distributed throughout it.

54

55 The temporal dynamics of the i^{th} phytoplankton size class (Eq. 5) is regulated by
 56 phytoplankton growth, $\mu_i A_i$, and by several loss processes. The growth rate of phytoplankton,
 57 described by,

58

$$\mu_i = \mu_{\max_i}(S_i^A) \cdot E(T) \cdot U(N) \cdot H(I_Z), \quad (\text{S8})$$

59

60 is regulated by a size-dependent maximum growth rate term,

61

$$\mu_{\max_i}(S_i^A) = \beta_{\mu_{\max}} \cdot \left(\frac{S_i^A}{1 \mu\text{m}} \right)^{\alpha_{\mu_{\max}}}, \quad (\text{S9})$$

62

63 with S_i^A representing the size of the i^{th} phytoplankton size class (Fig. 2A), and three environmental
 64 factors: temperature T , nutrient N , and light I_Z . The temperature-regulated growth term, $E(T)$,

following Norberg (2004), a new feature considered in this model, is presented in the main article, section ‘Temperature dependences’. The nutrient uptake term follows a Monod function (Monod 1949),

$$U(N) = \frac{N}{K_N + N}, \quad (S10)$$

with N representing the inorganic nutrient concentration in the lake (Eq. 4) and K_N the half-saturation constant for nutrient.

The light-limited growth term depends on the daily MLD-averaged photosynthetic rate,

$$H(I_z) = \frac{1}{MLD} \int_{z=0}^{z=MLD} P(I_z) dz, \quad (S11)$$

and is calculated by integrating the photosynthesis-irradiance curve (Lewis and Smith, 1983),

$$P(I_z) = P_{max} \frac{\alpha_{P_I} \cdot I_z}{\sqrt{P_{max}^2 + (\alpha_{P_I} I_z)^2}}, \quad (S11a)$$

through depth z (Eq. 12a). In this equation, α_{P_I} is the initial slope of the curve, P_{max} is the maximum photosynthetic rate, and I_z is a function estimating the level of irradiance, I , at depth z , based on the Beer-Lambert’s law:

$$I_z = I_0 \cdot e^{-K_{par} \cdot z}. \quad (S11b)$$

K_{par} is the light attenuation coefficient due to water, and I_0 is the irradiance value at surface, i.e., the observed PAR prescribed as forcing to the model. To optimize numerical performance, $H(I_z)$ is computed based on an analytical approximation proposed by Anderson et al. (2015):

$$H(I_z) = \frac{P_{max}}{K_{par} MLD} \log \left(\frac{\alpha_{P_I} I_0 + \sqrt{P_{max}^2 + (\alpha_{P_I} I_0)^2}}{\alpha_{P_I} I_z + \sqrt{P_{max}^2 + (\alpha_{P_I} I_z)^2}} \right). \quad (S11c)$$

Phytoplankton is lost due to natural mortality, ϕ_A , mixing, λ , and zooplankton grazing $\sum_i \sum_j G_{ij} Z_j$.

The temporal dynamics for the j^{th} zooplankton size class (Eq. 6) is regulated by net grazing, $\varepsilon\gamma\sum_i\sum_j G_{ij}Z_j$, which is the portion of grazed phytoplankton successfully ingested and assimilated, and several loss processes. The grazing rate of zooplankton is determined by,

$$G_{ij} = \psi_{\max_j}(S_j^Z) \cdot \psi(T) \cdot \left(\frac{\delta_{ij}(S_i^A, S_j^Z) A_i}{K_A + \sum_i \delta_{ij}(S_i^A, S_j^Z) A_i} \right). \quad (\text{S12})$$

The first factor describes the size-dependent maximum ingestion rate:

$$\psi_{\max_j}(S_j^Z) = \beta_{\psi_{\max}} \cdot S_j^{Z\alpha_{\psi_{\max}}}, \quad (\text{S13})$$

with S_j^Z representing the size of the j^{th} zooplankton size class (Fig. 2C). The second factor, $\psi(T)$, describes the temperature dependence of grazing and is presented in detail in the main article, section ‘Temperature dependences’ (Eq. 13A). The third factor of the grazing formulation comprises a half-saturation constant for phytoplankton intake, K_A , and the grazing preference for a specific range of phytoplankton size classes,

$$\delta_{ij}(S_i^A, S_j^Z) = e^{-\left[\frac{\log_{10}(S_i^A) - \log_{10}(P_{\text{opt}_j}(S_j^Z))}{\theta_j} \right]^2}. \quad (\text{S14})$$

The grazing preference is characterized by a Gaussian curve over a specific range of phytoplankton size classes. The median and the width of this Gaussian are determined by, respectively, the optimal phytoplankton size, $A_{\text{opt}_j}(S_j^Z)$, and the level of selectivity of the grazer θ_j (Fig. 2B). Grazers of different sizes are characterized by different optimal phytoplankton sizes following an allometric scaling as:

$$A_{\text{opt}_j}(S_j^Z) = \beta_{P_{\text{opt}}} \cdot S_j^{Z\alpha_{A_{\text{opt}}}}. \quad (\text{S15})$$

This function is empirically derived from the data of Hansen et al. (1997), reflecting the fact that large zooplankton prefer large phytoplankton cells (Fig. 2D).

116 The loss processes for zooplankton include natural mortality, ϕ_z , mixing, λ_z , and higher order
117 predation, $\eta_z Z_j$ (Eq. 6). The portion of grazed phytoplankton that is not ingested due to sloppy
118 feeding, $(1-\gamma)\varepsilon\sum_i\sum_j G_{ij} Z_j$, flows into the detritus pool (Eq. 7). The detritus pool acts as a nutrient
119 recycling station by remineralizing dead plankton and non-ingested materials from grazing. The
120 portion that is ingested but not assimilated, $(1-\gamma)\varepsilon\sum_i\sum_j G_{ij} Z_j$, is excreted and quickly remineralized
121 into the nutrient pool (Eq. 4).

122

123

124

125

126 **Supporting Information S2 – Environmental forcing**

127 As forcing to the model, we considered three environmental variables that are specific to Greifensee,
128 namely water temperature, photosynthetically active radiance (PAR), and mixed layer depth (MLD).

129

130 *Data source*

131 These forcing functions are derived from various time-series data, and the data cover different time
132 periods. Water temperatures are obtained from the CTD profiles as daily average epilimnetic
133 temperature from 2019 to 2022, measured by the Swiss Federal Institute of Aquatic Science and
134 Technology, Eawag. The PAR data is converted from solar radiance data collected from a weather
135 station next to Greifensee, measured daily by MeteoSwiss for the same period as the water
136 temperature. The PAR conversion is presented in the next paragraph. Finally, we obtained MLD data
137 from Pomati et al. (2020). These are monthly data collected at Greifensee from 1972 to 2015 (n=519).
138 To match the time step of the model (one day), we obtained daily values by interpolating the monthly
139 averages for MLD (see below).

140

141 *Unit conversion of PAR*

142 PAR values for plankton models are often obtained from light measurements, and light can be
143 expressed as photon flux density (i.e. $\text{Ein m}^{-2} \text{d}^{-1}$ / $\text{mol quanta m}^{-2} \text{s}^{-1}$), or as energy flow density (i.e.
144 W m^{-2}) (Lampert and Sommer, 2007). The photosynthetic processes of modelled phytoplankton are
145 often expressed in $\text{Ein m}^{-2} \text{d}^{-1}$ or $\text{mol quanta m}^{-2} \text{s}^{-1}$, whereas solar irradiance is commonly measured
146 in W m^{-2} . First, we calculate the amount of PAR from the solar irradiance in terms of net short-wave
147 solar irradiance, because not all wavelengths of the light spectrum are usable by photosynthesis. Only
148 wavelengths in the range 0.4-0.7 μm are suitable for most phytoplankton to perform photosynthesis,
149 and PAR is often considered in the range 0.41 to 0.50 of the net short-wave solar irradiance (Howell
150 et al. 1983; Tsubo and Walker 2005). The specific value depends on climatic conditions and differs
151 geographically. We adopt an intermediate value of 0.45 (Table S1, S2). After obtaining PAR in units
152 of W m^{-2} , we transformed these units into $\text{Ein m}^{-2} \text{d}^{-1}$. From Lampert and Sommer (2007), an
153 approximated conversion from W m^{-2} to $\mu\text{Ein m}^{-2} \text{s}^{-1}$ to ranges from 4 to 5. We adopt the value of 5
154 (Table S1, S2).

155

156 *Derivations of forcing function*

157 Based on the monthly values of the three sets of environmental data, we derive corresponding annual
158 forcing functions for water temperature, PAR, and MLD (Fig. 3). The water temperature is obtained
159 from CTD profiles as daily average epilimnetic temperatures from March 2019 to December 2022.
160 After obtaining monthly means from the daily averages over a one-year period, we interpolated the

161 monthly values to derive the forcing function for water temperature. The PAR data exhibits a wide
162 range of variability in the annual composite (Fig. 3), a common feature for irradiance data due to high
163 inter-annual or daily variabilities driven by cloud cover. We obtained the monthly means based on
164 the 75-percentile of the daily averages over a one-year period and interpolated the monthly values to
165 derive the forcing function for PAR (Fig. 3). The MLD data are retrieved from a monthly time-series
166 over 40 years (1972-2015) from Pomati et al. (2020). We manually derived twelve monthly data
167 points that capture the changes in the mixed layer depth and interpolate the monthly data into daily
168 values. The interpolations of the environmental data were performed using piecewise polynomials
169 (i.e. Spline) in the *scipy* package (Python 3.7.x) (Virtanen et al. 2020). The code for generating the
170 function is available on Github (<https://github.com/systemsecologygroup/TempSizeMod>). The
171 environmental data points and the generated forcing functions used as input to the model are shown
172 in Fig. 3.

173
174
175
176

177 **Supporting Information S3 – Conversions of observations and model results**

178 We used two raw plankton size datasets that are obtained from a field-deployed imaging device dated
179 from March 2019 to December 2022 (available in Supplementary Data,
180 doi:10.5281/zenodo.11390345). The plankton data for Greifensee, collected by the dual-
181 magnification dark-field imaging microscope (Orenstein et al. 2020), contain plankton abundance in
182 terms of ROI/second (Regions Of Interest per second) and plankton size estimates in terms of bio-
183 area (mm^2), minimum axis length (mm), and maximum axis length (mm). ROI/sec is a pseudo-unit
184 derived from the statistical output of each detected and processed object by the imaging microscope
185 that converts ROIs into image mosaics. In general, this unit is analogous to object counts for both
186 unicellular and colonial phytoplankton and zooplankton individuals (Merz et al. 2021). To be
187 comparable with model outputs, these data were converted into abundances (individual L^{-1}) based on
188 laboratory calibration functions (Merz et al. 2021). After that, the abundances were converted into
189 community biomass concentration ($\mu\text{M P}$), based on the size data of phytoplankton and zooplankton.
190 These conversions are explained in the following sub-sections and detailed equations are given in
191 Table S1.

192

193 **S2.1 Plankton abundances**

194 To obtain plankton abundances, we adopted the two calibrated functions derived by Merz et al. (2021)
195 and presented in Fig. S1. One equation was derived for the magnifier 5p0x (for smaller objects,
196 between 10 and 150 μm), whereas another was for the magnifier 0p5x (for larger objects, between
197 100 μm and 1 cm). We applied the equations to the plankton data according to the magnifier that
198 produced that data (Table S1). The phytoplankton species identified by the 5p0x magnifier are centric
199 diatom, cryptophyceae, rhodomonas, and chlorophytes, whereas those identified by the 0p5x
200 magnifier are dinobryon and uroglena. The zooplankton species captured from 5p0x and 0p5x are,
201 respectively, ciliates and daphnia.

202

203

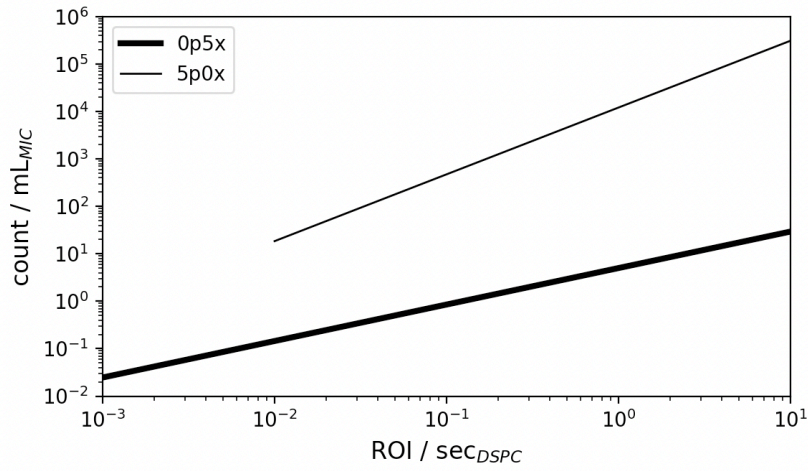


Fig. S1 Laboratory calibration of density estimated by microscopic count (count/mL) against density estimated by the plankton camera (ROI/sec). The functions are replotted from Merz et al. (2021): 5p0x ($r^2 = 0.889$) and 0p5x ($r^2 = 0.726$), Table S1.

S2.2 Phosphorus-equivalent phytoplankton biovolume

We transformed the bio-area (mm^2) estimates of phytoplankton to biovolume based on the relationship established by Ryabov et al. (2021), followed by calculating the phosphorus content using an allometric scaling function (Fig. S2) established by Hillebrand et al. (2022). The equations are shown in Table S1.

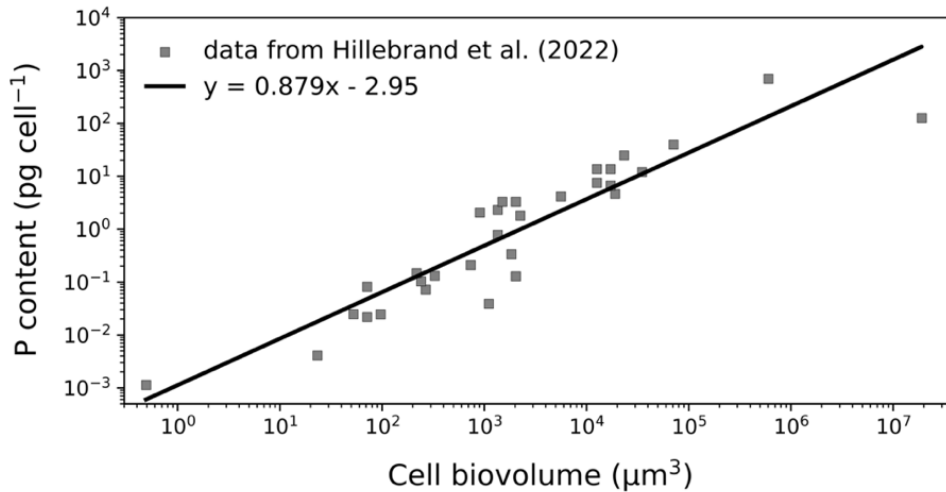


Fig. S2 The scaling function of phosphate content per cell in relation to phytoplankton biovolume (μm^3). The data (grey squares) are from Hillebrand et al. (2022) and the function (solid dark line) is a linear regression on the original data ($r^2 = 0.924$).

S2.3 Phosphorus-equivalent zooplankton biovolume

S2.3.1 Ciliates and Daphnia

The biovolume of ciliates was approximated assuming that the general shape of ciliates is a standard ellipsoid, following Hillebrand et al. (1999). We use the major and minor axis length data of the

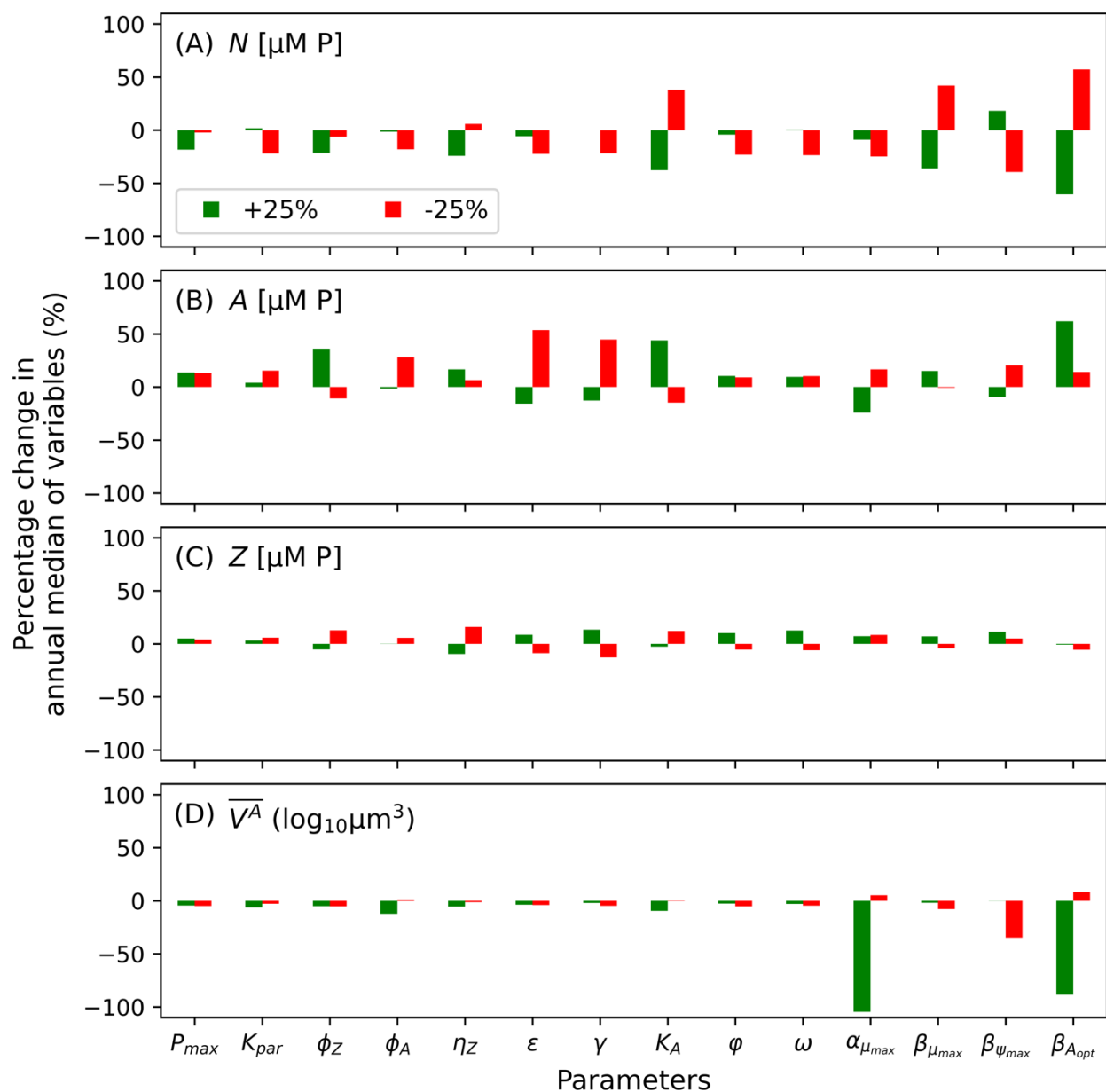
224 ciliates to represent the principal diameters of the ellipsoidal ciliate and assume that the third
225 dimension of the ellipsoidal ciliate is equal to the minor axis length (Table S1). We obtain the carbon
226 content per ciliate individual in relation to size according to the size-scaling relationship of Putt and
227 Stoecker (1989) and the stoichiometric transformation of carbon to phosphorus is based on a range
228 within 80-120 measured for a freshwater ciliates *Euplotes sp.* (Golz et al. 2015). We used the major
229 axis length data of daphnia to estimate their carbon content based on a body length to weight
230 relationship in Schalau et al. (2008) and considered a correction factor of 0.75 from the major axis
231 length to body length. The stoichiometric conversion of carbon to phosphorus is based on a ranges
232 from 80 to 385 in previous modelling studies (reviewed in Khattak et al. 2018), Table S1. The detailed
233 conversion is shown in Table S1, and the values of the conversion coefficients are given in Table S2.
234 Those values were manually optimized by comparing and matching the plankton abundances
235 obtained from the observations with the abundances produced by the model.
236

Supporting Information S4 – Sensitivity analysis on standard model parameters

We conducted a sensitivity analysis to evaluate the uncertainty of those parameters that could not be directly estimated from Greifensee data (Table 1). A total of 14 parameters were tested, except the thermal traits that were tested in a different sensitivity analysis (Fig. 4) and the parameters that were already calibrated by matching model results to observations (Fig. S5). We varied the 14 parameters by $\pm 25\%$ from standard value (Table 1) and quantified the percentage change on the annual median for four model variables, nutrient concentrations, community-integrated phytoplankton and zooplankton biomass, and community-mean phytoplankton biovolume. This sensitivity test is simulated for the standard model under the default physical forcing.

The result shows that nutrient concentration and phytoplankton biomass are generally sensitive to the parameter tested (Fig. S3A, B). Zooplankton biomass and community-mean phytoplankton biovolume are generally not very sensitive to the parameter tested (Fig. S3C, D). Nutrient concentrations are most sensitive (affect nutrient values by max. $\pm 60\%$) to the slope and intercept of allometric relationships for phytoplankton growth ($\alpha_{\mu_{max}}$, $\beta_{\mu_{max}}$), zooplankton grazing ($\beta_{\psi_{max}}$), and optimal phytoplankton size for zooplankton ($\beta_{A_{opt}}$), Fig. S3A. This suggests that nutrient concentration is sensitive to growth and grazing parameters that vary with size-dependencies. Phytoplankton concentrations are more sensitive (altered by max. $\pm 60\%$) to parameters that control grazing, including zooplankton mortality (ϕ_z), fraction of grazed material that is not ingested, i.e., sloppy feeding (ε), fraction of grazed material that is assimilated into zooplankton growth (γ), half-saturation constant for grazing (K_A), and the intercept of allometric relationship of optimal phytoplankton size for zooplankton grazing ($\beta_{A_{opt}}$), Fig. S3B. Other parameters produced weaker effects to the annual median value of nutrient and phytoplankton biomass, i.e., within $\pm 30\%$ (Table S4).

Our result shows moderate changes in zooplankton (within about $\pm 15\%$), Table S4. The community-mean phytoplankton biovolume is insensitive to most parameters (within $\pm 10\%$), except to the $+25\%$ slope of allometric relationships for maximum phytoplankton growth ($\alpha_{\mu_{max}}$) and the $+25\%$ intercept of allometric relationships of optimal phytoplankton size for zooplankton grazing ($\beta_{A_{opt}}$), Fig. S3D. Increasing these two parameters favours small phytoplankton by decreasing the growth ability of larger phytoplankton and by weakening grazing on small phytoplankton (grazing shifts to target larger phytoplankton). Both parameters produce a strong decline in mean phytoplankton biovolume (Fig. S3D).



272

273 **Fig. S3** Results for the sensitivity analysis on fourteen model parameters. These are
274 parameters that were neither examined in sensitivity analyses nor calibrated with observations. The
275 result shows changes in annual median of four model variables, (A) nutrient concentration in the
276 epilimnion, (B) phytoplankton biomass, (C) zooplankton biomass, and (D) phytoplankton mean
277 biovolume, by +25% (green bars) and -25% (red bars) defaulted parameter values (Table 1). The
278 percentage change values are reported in Table S4.

279 **Supporting Information S5 – Sensitivity test on alterations of lake stratification**

280 In a warming world, higher temperature and longer stratification periods are factors (Woolway et al.
281 2020, 2021) that can affect nutrient transport (Behrenfeld et al. 2006), plankton biomass, and plankton
282 phenology (Winder and Sommer 2012). Earlier onsets of stratification and longer periods of
283 stratification are expected to occur in multiple temperate lakes across the Northern Hemisphere
284 (Winder and Schindler 2004; Shatwell et al. 2019; Woolway et al. 2021).

285

286 Since our ecological projection did not consider any changes in the lake stratification, we
287 performed a sensitivity test varying the lake mixing dynamics under the most severe climate change
288 scenario, RCP 8.5. Lake stratification in our model is governed by two features, variations of the
289 mixed layer depth (MLD) in combination with a cross-thermocline diffusive mixing, captured by a
290 constant parameter (ω). Based on the estimation of the median change in stratification period by
291 Woolway et al. (2021), we created a prolonged lake stratification that starts 22 days earlier and ends
292 11 days later (Fig. S4A). This leads to a longer summer stratification, by 33 days. In another case, we
293 reduced the diffusive mixing coefficient by 50%. This parameter determines the strength of diffusive
294 mixing between hypolimnion and epilimnion.

295

296 The model output shows that, these variations produce effects to all variables. The more
297 prominent effects include a lower level of nutrient concentration throughout the year (Fig. S4B) and
298 an earlier zooplankton spring peak under a prolonged stratification (Fig. S4D). A weaker diffusive
299 mixing produces a lower level of nutrient accumulation (30% reduction in annual median) in summer
300 compared to the standard mixing dynamics (Fig. S4B, inset). This is likely due to the reduced amount
301 of nutrient recharged from the hypolimnion. Additionally, in our model, an earlier lake stratification
302 leads to earlier development of zooplankton peak in spring. That is because the zooplankton peak is
303 sensitive to the start of stratification (since the shallowing of mixed layer depth increases biomass
304 concentration, Eq. S1), but they are not affected by changes in diffusive mixing. The alteration in lake
305 stratification produces only minor changes to the biomass dynamics and community mean biovolume
306 of phytoplankton (Fig. S4C, E).

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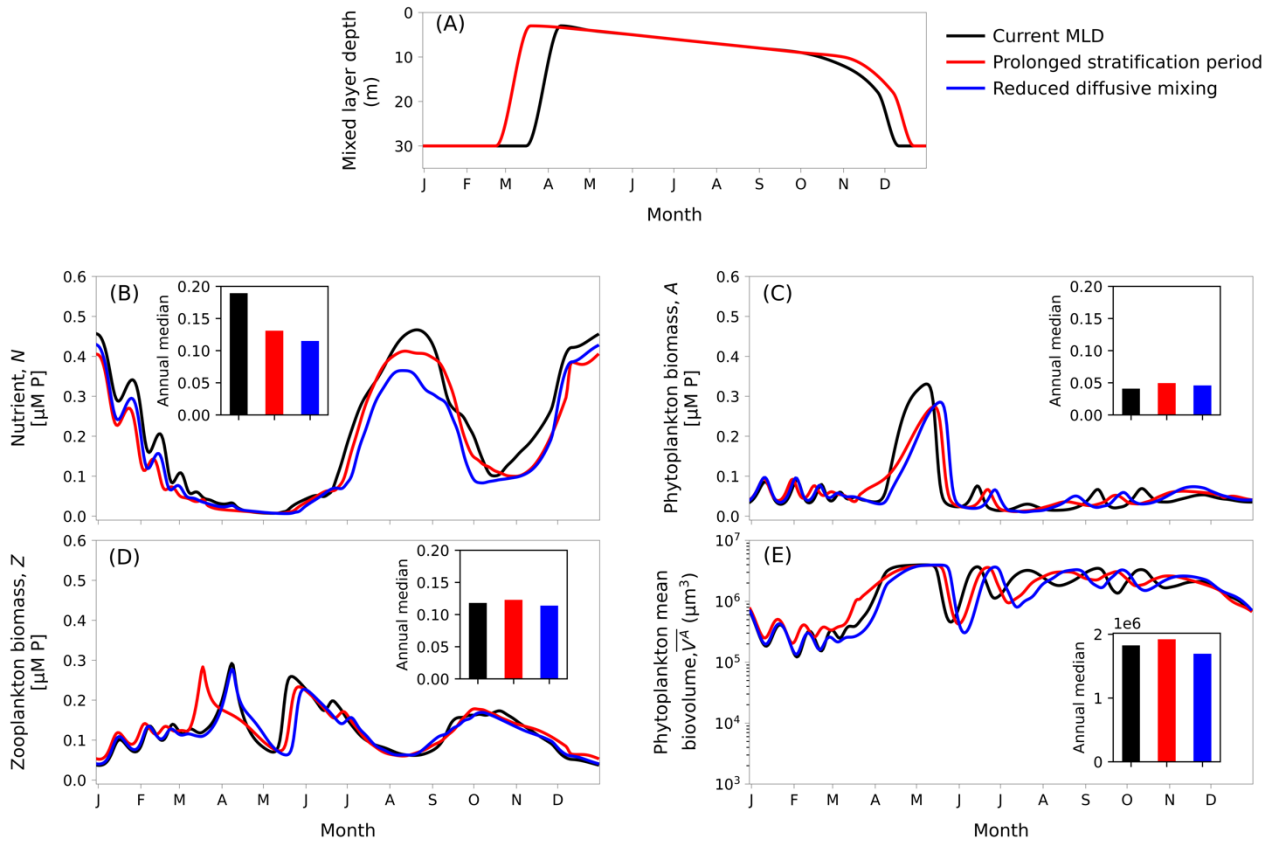


Fig. S4 Results of sensitivity test on alterations of lake stratification compared to the standard mixing regime under the most severe climate change scenario (RCP 8.5, water temperature increases by 4.2 °C throughout the year). (A) Standard mixed layer depth (black), future mixed layer depth produced by (1) a prolonged stratification period (red) and (2) a weakened diffusive mixing between hypolimnion and epilimnion (blue). The results show, for the three mixing regimes, temporal dynamics of (B) nutrient concentration in the epilimnion, (C) phytoplankton biomass, (D) zooplankton biomass, and (E) phytoplankton mean biovolume. The insets show the annual median value for different alterations. The model is run to a steady annual cycle over a period of 10 years with repeated annual forcing, the results shown are those of the 10th year.

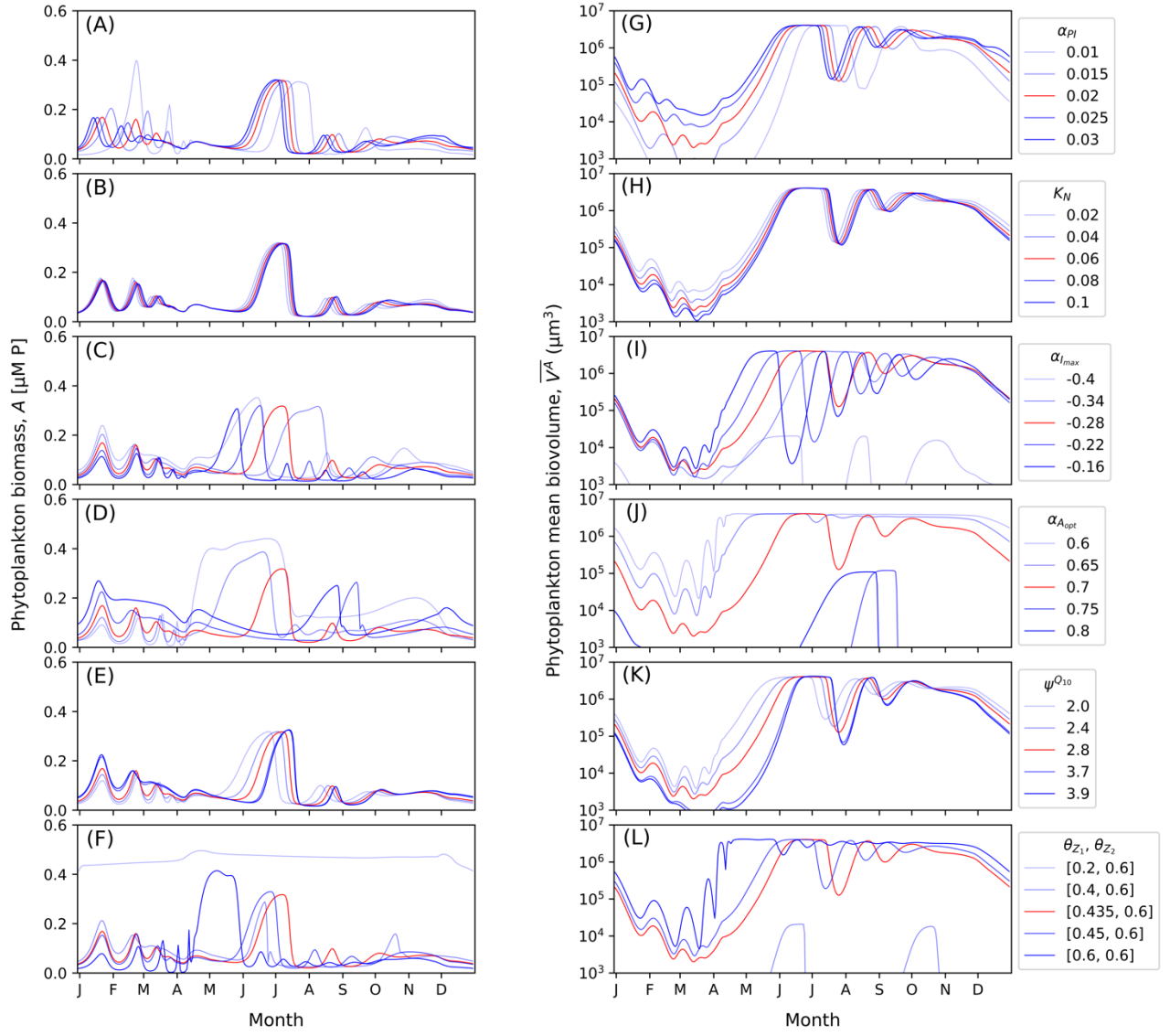
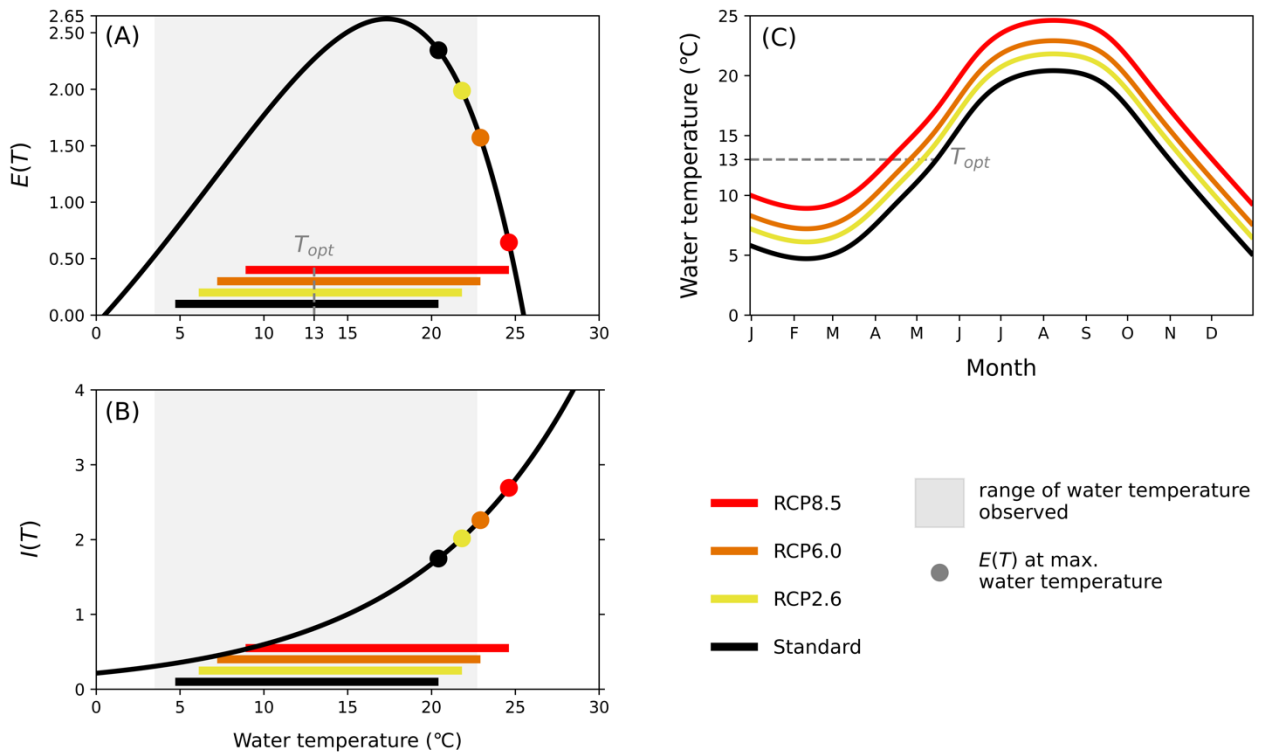


Fig. S5 Selected results for the calibrated parameters. (A, G) initial slope of P-I curve, α_{PI} ; (B, H) half saturation constant for nutrient, K_N ; (C, I) slope of the allometric relationship for maximum ingestion rate, $\alpha_{I_{max}}$; (D, J) slope of the allometric relationship for optimal phytoplankton size, $\alpha_{A_{opt}}$; (E, K) Q_{10} coefficient for maximum ingestion rate, $\psi^{Q_{10}}$; and (F, L) size-grazing breadth for Z_1 and Z_2 , $\theta_j^{Z_1}$ & $\theta_j^{Z_2}$. The results for (A-E) phytoplankton biomass concentrations, A , and (F-J) phytoplankton community mean biovolume, $\overline{V^A}$. The red line indicates the value used for the standard model run.



332

333 **Fig. S6** Temperature dependence curves for (A) phytoplankton growth, $E(T)$, and (B)

334 zooplankton grazing, $I(T)$, for different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0

335 (orange), and RCP 8.5 (red). (C) Optimal temperature for growth, T_{opt} . The standard range of water

336 temperature is represented by the black line. The climate change scenarios simulate an increase in the

337 standard range of water temperature by, respectively, 1.4 °C (RCP 2.6, yellow bar), 2.5 °C (RCP 6.0,

338 orange bar), and 4.2 °C (RCP 8.5, red bar). The circles (A, B) indicate the level of temperature

339 dependence at the hottest day of the year from different scenarios and show that phytoplankton growth

340 is becoming increasingly suboptimal while zooplankton grazing intensifies.

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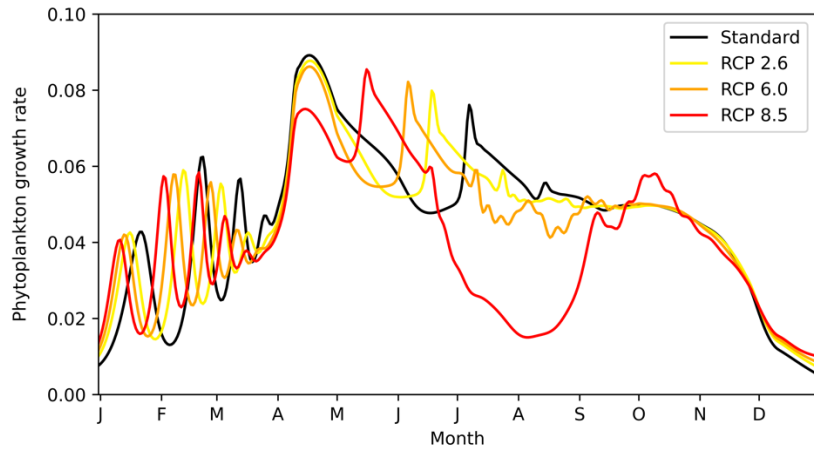


Fig. S7 Phytoplankton growth rates, $\sum \mu_i$ (Eq. S8), in different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red). It shows a pronounced decline in summer phytoplankton growth under RCP 8.5.

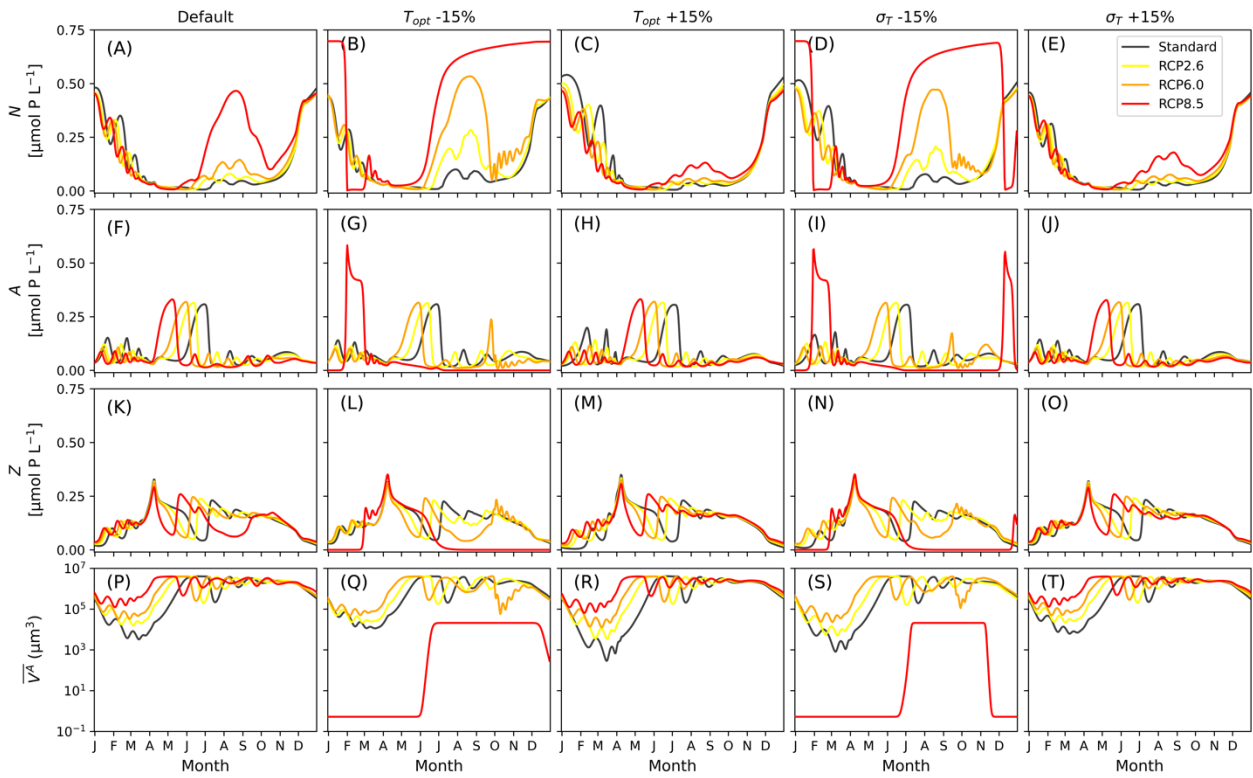


Fig. S8 Results of the sensitivity analysis ($\pm 15\%$) related to the two thermal traits, T_{opt} and σ_T . The result shows the temporal dynamics of four model variables: nutrient concentration, N (A-E); phytoplankton biomass concentration, A (F-J); zooplankton biomass concentration, Z (K-O); and phytoplankton community mean biovolume, $\bar{V}^A$ (P-T), under the different scenarios: standard (dark), RCP 2.6 (yellow), RCP 6.0 (orange), and RCP 8.5 (red).

356 **Table S1** Transformation of model and data output with physical, stoichiometric, and size-scaling coefficient. The references for the equations are provided below the
 357 equations and those for specific coefficient are given in Table S2

		Solar irradiance (W m ⁻²)	Abundance data		Phytoplankton size		Ciliates Minimum and maximum axis length (mm)	Ciliates carbon content (pg C ind ⁻¹)	Daphnia carbon content (µg C ind ⁻¹)
			ROI/sec (<100 µm)	ROI/sec (>100 µm)	Bio-area, A (µm ²)	Modeled ESD (µm)			
Physical	Net Short-wave solar irradiance i.e. light usable for photo- synthesis (W m ⁻²)	0.45 (Table S2)							
	Photosynthetically Active radiance (E m ⁻² d ⁻¹)	$\frac{5 \cdot 86400}{10^6}$ (Table S2)							
Abundance	Phytoplankton (ind L ⁻¹)		$10^{\frac{2.9}{0.71}} \cdot (10^{ROI/sec})^{\frac{1}{0.71}}$ Merz et al. (2021)						
	Zooplankton (ind L ⁻¹)		$10^{\frac{0.91}{1.3}} \cdot (10^{ROI/sec})^{\frac{1}{1.3}}$ Merz et al. (2021)						
Phytoplankton	Biovolume, V (µm ³)				$A^{\frac{1}{(0.68 \cdot 0.25)}}$ Ryabov et al. (2021)	$\frac{4}{3} \cdot \pi \cdot \left(\frac{ESD}{2}\right)^3$ Hillebrand et al. (1999)			
Zooplankton	Ciliates biovolume, V (µm ²)						$\frac{\pi}{6} \cdot a \cdot b \cdot h \cdot 10^3$ Hillebrand et al. (1999)	$10^{0.12} \cdot V^{0.19}$ Putt and Stoecker (1989)	
	Daphnia maximum axis length, L (mm)								$1.6 \cdot L^3$ Schalau et al. (2008)
Stoichiometry	Phosphate content per individual cell (pg P ind ⁻¹)				$10^{-2.95} \cdot V^{0.88}$ (Hillebrand et al., 2022)				
	Phosphorus- equivalent biomass concentration (µmol P ind ⁻¹)				$\frac{1}{mw\ P} \cdot 10^{-6}$ (Table S2)			$\frac{1}{mw\ C} \cdot \frac{1}{C:P_{Ci}} \cdot 10^{-6}$ (Table S2)	$\frac{1}{mw\ C} \cdot \frac{1}{C:P_{Da}}$ (Table S2)

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359 **Table S2** Parameters for physical and stoichiometric conversion and their descriptions, standard values,
 360 parameter range, units, and sources for the parameter or parameter range. Those that are provided with a range
 361 was calibrated by manually comparing the range of biomass values obtained from the data with those from the
 362 model.

Parameter	Description	Standard Value	Parameter range	Units	Source for parameter
-	Solar irradiance to net short-wave solar irradiance	0.45	0.41-0.5	W m ⁻²	Howell et al. (1983); Tsubo and Walker (2005)
-	Conversion of energy flow density to photon flux density	5	4-5	μE m ⁻² s ⁻¹ (mol P) ⁻¹	Lampert and Sommer (2007)
mw P	Molar weight of phosphorus	31.97	-	μg P	Standard unit
mw C	Molar weight of carbon	12.01	-	μg C	Standard unit
C: P _{Ci}	C:P ratio for ciliates	100	80-120	mol C (mol P) ⁻¹	Golz et al. (2015)
C: P _{Da}	C:P ratio for daphnia	350	80-385	mol C (mol P) ⁻¹	Khattak et al. (2018)

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366 **Table S3** Information for the increase in lake water surface temperature (LWST) for future projections (i.e.
 367 year 2099 or 2100) and the corresponding sources.

RCP scenarios	Value used for projections	Range of changes in LWST (°C)	Location	References
RCP 2.6	+1.4°C	1.36	Lake Erken, Sweden	Ayala et al. (2020) ¹
(low-emission scenario)		1.4	10 Polish lakes	Czernecki and Ptak (2018) ²
		0.09 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020) ³
		0.51 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023) ⁴
RCP 6.0	+2.5°C	2.31	Lake Erken, Sweden	Ayala et al. (2020)
(medium-high-emission scenario)		1.8 – 2.7	The Great Lake region	Xue et al. (2022) ⁵
		0.32 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020)
		1.41 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023)
RCP 8.5	+4.2°C	4.2	10 Polish lakes	Czernecki and Ptak (2018)
(high-emission scenario)		4.1 – 5.0	The Great Lake region	Xue et al. (2022)
		0.5 decade ⁻¹	Rappbode Reservoir, Germany	Mi et al. (2020)
		4.51 century ⁻¹	Lake Kozjak, Croatia	Šarović and Klaić (2023)

368 ¹ Ayala et al. (2020) adopts the one-dimensional hydrodynamic model, GOTM, to simulate the lake surface water temperature in
 369 future scenarios that are projected by the GCMs from ISIMIP scheme. Their long-term simulations were forced with daily or hourly
 370 meteorological data.
 371 ² Czernecki and Ptak (2017) simulates future LWST using empirical-statistical downscaling modelling techniques, based on
 372 simulations of 33 atmosphere-ocean coupled GCMs obtained from the CMIP5 project.
 373 ³ Mi et al. (2020) uses a well-developed hydrodynamic model and consider a set of reanalyzed daily meteorological input (bias-
 374 corrected for ISIMIP).
 375 ⁴ Šarović et al. (2023) uses a one-dimension energy budget model to simulate changes in mean lake water temperature, based on
 376 projected meteorological data (i.e. CORDEX) taken from the World Climate Research Programme.
 377 ⁵ Xue et al. (2022) couples a regional climate model with a three-dimensional hydrodynamic lake model, GLARM, using GCMs
 378 obtained from the CMIP5 project.
 379

381 **Table S4** Percentage change on the annual median value of four model variables: nutrient concentration in
 382 the epilimnion, N , phytoplankton biomass, P , zooplankton biomass, Z , and phytoplankton mean biovolume, $\overline{V^A}$,
 383 under $\pm 25\%$ change in fourteen model parameters for the standard model run.

Sensitivity of variables	N [$\mu\text{M P}$]		P [$\mu\text{M P}$]		Z [$\mu\text{M P}$]		$\overline{V^A}$ ($\log_{10} \mu\text{m}^3$)	
% Change in parameter values	+25%	-25%	+25%	-25%	+25%	-25%	+25%	-25%
P_{max}	-18.31	-2.31	13.68	13.38	5.06	4.1	-4.48	-4.93
K_{par}	1.74	-21.85	4.05	15.44	3.15	5.81	-6.11	-2.85
ϕ_A	-21.68	-6.32	36.17	-10.73	-5.16	12.7	-5.12	-5.25
ϕ_Z	-1.53	-18.06	-1.45	28.14	0.18	5.61	-12.33	0.97
η_Z	-24.25	5.87	16.54	6.52	-9.53	15.89	-5.49	-1.28
ε	-5.83	-22.43	-15.71	53.66	8.6	-8.74	-3.68	-3.96
γ	0.11	-21.7	-12.7	44.72	13.35	-12.78	-1.98	-4.79
K_A	-37.68	37.81	43.86	-14.7	-2.56	12.19	-9.61	0.37
φ	-4.31	-23.1	10.41	9.03	10.06	-5.42	-2.6	-5.31
ω	0.35	-23.68	9.53	10.33	12.62	-5.99	-2.95	-4.62
$\alpha_{\mu_{max}}$	-9.03	-24.79	-24.05	16.65	7.25	8.42	-104.65	5.13
$\beta_{\mu_{max}}$	-36.06	42.03	15.11	-0.66	6.99	-4.0	-1.87	-7.88
$\beta_{\psi_{max}}$	18.1	-39.41	-9.16	20.39	11.55	5.1	0.09	-34.71
$\beta_{A_{opt}}$	-60.51	57.14	61.83	14.11	-0.77	-5.48	-88.48	8.12

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